

From Defect Detection to Flight Clearance: a Bayesian SHM Stack for Reusable Launch Vehicles

Stage 0–3 implemented and verified on CFRP thermoelastic fields

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Motivation: reusable rockets change the question

Expendable: “will it survive one flight?”

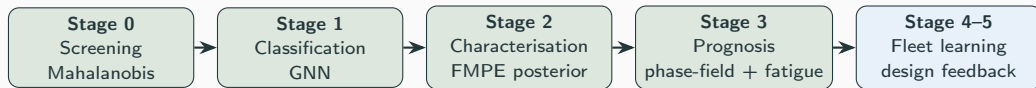
Reusable: “is this vehicle OK to fly *again* — and how many flights remain?”

- Same vehicle re-inspected after every flight
⇒ *fixed geometry, fixed individual* is the norm
- SpaceX (empirically): condition-based monitoring, staged certification 20 → 40 flights, fleet-leader strategy (B1067: 35 flights, 06/2026)

This work

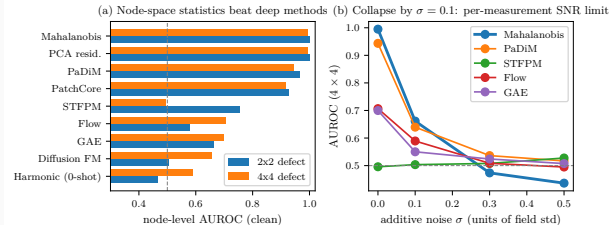
Formalise that practice as a measurable, Bayesian, physics-based pipeline — applicable from the *first* article.

The Stage 0–5 stack



- **Green = implemented and tested** (2026-06-11, two repos, all committed)
- One data contract end-to-end: posterior $\theta = (c_x, c_y, \text{layer}, \text{size})$ flows from Stage 2 directly into Stage 3

Stage 0 — a 30-line statistic beats eight learned methods



Node-level AUROC (clean):

- per-node Mahalanobis: **0.999 / 0.995** (2x2/4x4)
- deep methods: 0.50–0.75

SNR limit

All methods collapse at noise

$\sigma=0.1$:

defect amplitude $< 0.1\sigma$. Lock-in IR averages 10^3 cycles $\Rightarrow$ requirement, not blocker.

Why the simple method wins here — and why GNN still matters

Fixed geometry = template statistics

Re-inspecting the *same* vehicle: healthy manifold is a point cloud around one template. Per-node (μ_i, σ_i) is the correct inductive bias.

Complementary weaknesses

	Mahalanobis	GNN
unknown defect size	open	collapses
noise (12%)	dies	robust
semantic class	—	only option
new airframe	needs refs	transferable

⇒ statistics dominate as a vehicle accumulates flight history; learned models bootstrap *new* airframes and repairs.

Stage 2 — calibrated Bayesian defect characterisation (FMPE)

Flow-matching posterior estimation, trained on the *existing* FEM set (20k pairs, zero new simulations):

parameter	RMSE	R^2	90%-coverage
position c_x	0.055	0.78	0.944
position c_y	0.021	0.83	0.940
size (3-class)	—	0.73 (acc. 0.82)	0.894
layer (depth)	2.98	0.63	0.870

- Coverage $\approx$ target 0.90 $\Rightarrow$ **calibrated** posteriors, safe to propagate downstream
- Depth weakly identifiable — a physics limit of surface thermoelastic measurement, reported honestly

Stage 3 — phase-field prognosis with true fatigue

Model (12/12 tests green):

- multi-ply anisotropic AT2, $A = I + \beta a \otimes a$ per ply
- interface bands ($G_c^{int}/G_c^{ply}=0.4$, DCB-anchored)
- Carrara-2020 fatigue: history $\bar{\alpha}$ degrades $G_c \Rightarrow$ Wöhler-like growth under repeated sub-critical loads

Verified physics:

- fiber at $30^\circ \rightarrow$ crack deflects to 23.3°
- low- G_c interface captures crack (73 vs 0 cells)
- $P(\text{growth})$ monotone in load:
 $0 \rightarrow \frac{2}{3} \rightarrow 1$

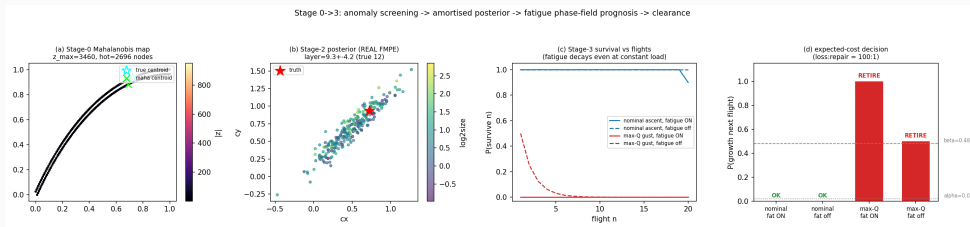
Flight clearance — end-to-end on a real test sample

Stage 0→3 run: held-out 4x4 field → Mahalanobis map → *trained* FMPE posterior → fatigue-aware clearance (20-flight horizon).

load case	load	fatigue	$P(\text{grow})$	$\mathbb{E}[\text{flights}]$	decision
nominal ascent	0.06	ON	0.00	19.9	OK
nominal ascent	0.06	off	0.00	11.0	OK
max-Q gust	0.10	ON	1.00	0.0	RETIRE
max-Q gust	0.10	off	0.50	1.0	RETIRE

- fatigue *accumulates across flights* — $P(\text{survive } 20)$ now decays even at constant sub-critical load (no i.i.d. shortcut)
- thresholds from expected cost (vehicle:repair = 100:1 $\Rightarrow \alpha=0.02, \beta=0.48$)

End-to-end, one figure: detect $\rightarrow$ characterise $\rightarrow$ clear



Stage 0 anomaly map $\cdot$ Stage 2 posterior $\cdot P(\text{survive } n)$ with fatigue $\cdot$ decision — single held-out CFRP field.

Outlook: Stage 4–5 and deployment

- **Stage 4 — fleet learning:** hierarchical Bayes over tail numbers; formalises the SpaceX fleet-leader practice (lead vehicle = likelihood, fleet = prior update)
- **Stage 5 — design feedback:** fleet damage statistics → layup/reinforcement optimisation (BO)
- **Sim-to-real:** flow-matching bridge ready for first measured full-field data (per-sample normalisation needs no healthy reference)
- **Targets:** JAXA reusable-vehicle programmes; journal paper (anomaly benchmark + calibrated FMPE + fatigue prognosis)

One sentence

SpaceX learned this loop empirically over a decade — we formalise it so it works from the first inspection of the first article.